

# Quantum Geometric Observables for Financial Regime Detection

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## Abstract

Classical regime detection methods operate in flat Euclidean feature spaces, relying on distributional assumptions that are violated during the very transitions they seek to detect. We construct a learned Riemannian manifold for financial markets by lifting time series into a projective Hilbert space  $\mathbb{CP}^{n-1}$  via an error Hamiltonian and extracting the induced geometric structure: quantum metric tensor, Berry curvature, spectral gap, and quantum Fisher information (QFI) matrix. This geometry induces a family of scalar observables—Berry phase rate, QFI pseudo-determinant, rolling Chern numbers, multi-lag fidelity, and metric condition number—that summarize the local and global structure of the market manifold at each point in time. In a head-to-head comparison of 16 methods across 12 historical crises (2007–2023), we show that: **(i)** Optuna-optimized fusion of the three top geometric observables achieves median Cohen’s  $d = 1.67$  across all crises—48% above the supervised Random Forest baseline ( $d = 1.13$ ) despite requiring no labeled training data; **(ii)** under a strictly causal protocol that prevents lookahead bias, individual geometric observables achieve  $d > 2.0$  (QFI  $d = 2.30$ , multi-lag fidelity  $d = 2.29$ ); **(iii)** the geometric framework generalizes across asset classes: all three methods maintain  $d > 1.0$  on five broad-market ETFs (SPY, QQQ, IWM, EFA, DIA) with Friedman  $\chi^2 = 35.0$ ,  $p < 0.0001$ ; and **(iv)** the geometric signal is complementary to supervised detection—on novel crises (SVB, COVID), QCML achieves  $d > 1.7$  where Random Forest scores  $d < 0.5$ , while a hybrid ensemble nearly doubles out-of-sample  $d$  relative to RF alone ( $d = 0.60$  vs.  $d = 0.31$ ). These findings establish the learned market manifold as a principled geometric framework whose topological and curvature properties encode regime structure, with greatest value precisely where supervised approaches are weakest.

**Keywords:** regime detection, quantum geometry, Chern number, Berry curvature, quantum metric, topological invariants, financial crises, quantum Fisher information.

## 1 Introduction

Detecting regime transitions is a central problem in quantitative finance. Market dynamics alternate between qualitatively different states—low-volatility trends, high-volatility drawdowns, liquidity crises—and the ability to identify transitions in real time has implications for risk management, portfolio construction, and hedging.

Classical approaches include Hidden Markov Models (HMMs) [1], cumulative-sum (CUSUM) statistics [2], rolling volatility breakpoints, and supervised classifiers such as Random Forests. While effective in specific settings, these methods share a common limitation: they operate entirely within a Euclidean feature space and rely on distributional assumptions (Gaussian mixtures, stationarity within regimes) that are frequently violated during the very transitions they seek to detect.

Our central thesis is that financial regimes correspond to distinct topological phases of a learned market manifold, and that regime transitions manifest as geometric singularities—curvature spikes, spectral gap closures, and changes in topological charge—of this manifold.

Why this geometric detour, rather than a simpler nonlinear method such as a kernel machine or neural network? Projective Hilbert space provides four structural advantages that no purely Euclidean approach can replicate. *First*, the Chern number  $c_1$  is a topological invariant and hence an integer; small perturbations to the data shift the Berry curvature smoothly but cannot change  $c_1$  until a genuine phase transition occurs, yielding a noise-robust classification that continuous anomaly scores cannot match. *Second*, the Berry curvature  $F_{ab}$  is gauge-invariant—it is unchanged under unitary rotations  $|\psi\rangle \mapsto e^{i\theta}|\psi\rangle$ —whereas kernel-based distances depend on the choice of coordinate representation. *Third*, the spectral gap  $\Delta(x)$  between ground and first excited state closes *continuously* before a topological transition, providing a principled leading indicator with no ad hoc threshold selection. *Fourth*, the non-commutativity of the Hermitian operators  $\{O_j\}$  encodes cross-feature interactions (e.g., momentum–volatility coupling) in the off-diagonal structure of  $H(x)$ , capturing higher-order dependencies that PCA-based methods, which diagonalize by construction, necessarily discard.

In this paper we construct a learned Riemannian manifold for financial markets and show that regime detection emerges as a natural consequence of its geometry. Inspired by the Quantum Cognition Machine Learning (QCML) framework [3], we *lift* the feature space  $\mathbb{R}^d$  into a projective Hilbert space  $\mathbb{CP}^{n-1}$  via a family of Hermitian operators. The ground state  $|\psi(x)\rangle$  of an error Hamiltonian  $H(x)$  defines a smooth map whose differential geometry inherits topological structure from quantum mechanics. Specifically:

1. The *quantum metric tensor*  $g_{ab}$  (the real part of the Fubini–Study metric pulled back to  $\mathbb{R}^d$ ) endows the data manifold with a natural Riemannian structure, replacing the ad hoc Euclidean metric.
2. The *Berry curvature*  $F_{ab}$  (the antisymmetric, imaginary part) [13] gives rise to topological invariants—*Chern numbers*—that are integers for closed manifolds and therefore intrinsically robust to continuous perturbations (noise).
3. Discontinuities in the rolling Chern number distinguish *topological* regime transitions ( $\Delta C \neq 0$ ) from extreme events within the same regime ( $\Delta C = 0$ ), a distinction inaccessible to purely statistical detectors.

## Contributions.

1. **The geometry.** We construct the full Riemannian and topological geometry of a financial data manifold via the QCML error Hamiltonian, including the quantum metric tensor, Berry curvature, Christoffel symbols, Ricci scalar, spectral gap, and quantum distances (Section 3).
2. **Geometric observables.** The geometry induces a family of scalar observables—Berry phase rate, QFI pseudo-determinant, Chern numbers, multi-lag fidelity—that summarize local and global manifold structure at each point in time (Section 4).
3. **Empirical validation.** We show that these observables change sharply around known regime transitions across 12 historical crises on five broad-market ETFs. Optimized fusion of the top three observables achieves median  $d = 1.67$ , exceeding the supervised Random Forest baseline ( $d = 1.13$ ) by 48%, and individual methods reach  $d > 2.0$  under strictly causal protocols (Section 6).

## 2 Background and Related Work

### 2.1 Classical Regime Detection

The regime-switching framework originates with Hamilton’s Markov-switching autoregression [1], in which a latent discrete state governs the parameters of an observed process. The two-state Gaussian HMM remains a workhorse, estimating  $P(\text{high-vol state} | r_t)$  via the Baum–Welch algorithm. CUSUM statistics [2] provide a nonparametric alternative, accumulating deviations  $S_t = \max(0, S_{t-1} + |r_t| - k)$  and flagging  $S_t > h$  as a changepoint. Rolling volatility z-scores are perhaps the simplest approach: one computes  $z_t = (\sigma_t^{\text{short}} - \mu_t^{\text{long}}) / \sigma_t^{\text{long}}$  and applies a threshold. Supervised classifiers (Random Forest, gradient-boosted trees) can achieve high in-sample accuracy but require labeled data and are prone to overfitting on rare events [4].

**Bayesian online changepoint detection.** Adams and MacKay [14] introduced Bayesian Online Changepoint Detection (BOCPD), which maintains a posterior distribution over the *run length*—the time since the last changepoint—and updates it recursively as new observations arrive. Extensions to multivariate settings and non-exponential-family likelihoods have been explored by Knoblauch and Damoulas [15]. While BOCPD provides principled uncertainty quantification over changepoint timing, it remains a distributional method operating in flat Euclidean space.

All of these methods treat features as elements of a flat Euclidean space. Our approach differs by constructing a learned Riemannian manifold whose curvature and topology encode structural information about the market state—and whose geometric observables, as we show, change sharply at regime boundaries.

### 2.2 Quantum-Inspired Machine Learning

The QCML framework [3] embeds classical data into quantum mechanical structures. Each data point  $x \in \mathbb{R}^d$  is associated with a Hamiltonian  $H(x)$  whose ground state  $|\psi(x)\rangle$  serves as a nonlinear feature map  $\mathbb{R}^d \rightarrow \mathbb{CP}^{n-1}$ . This is conceptually analogous to kernel methods—the Fubini–Study metric induces a positive-definite kernel  $K(x, y) = |\langle \psi(x) | \psi(y) \rangle|^2$ —but additionally provides geometric and topological structure absent from standard kernel machines.

### 2.3 Topological Data Analysis in Finance

Topological data analysis (TDA) uses persistent homology to detect early warning signals of market crashes. Gidea and Katz [16] computed Betti numbers from sliding-window point clouds of daily returns, showing that persistent homology norms spike  $\sim 5$  trading days before crashes. Gidea [17] extended this to financial correlation networks. Our approach differs in that we extract topology from a *learned* quantum-mechanical structure (Berry curvature and Chern numbers) rather than from point-cloud pairwise distances, providing a richer geometric structure (metric tensor, curvature, spectral gap) beyond topology alone.

### 2.4 Deep Learning Approaches to Regime Detection

Deep learning methods—including LSTMs [18], Temporal Convolutional Networks [19], Transformer-based architectures, and autoencoder-based anomaly detection [20]—have been applied to regime detection. While attention mechanisms can capture long-range dependencies, these approaches remain fundamentally supervised, requiring labeled data and tending to overfit on the small number of

historical crises available. Our geometric observables are unsupervised, detecting regime transitions as intrinsic geometric singularities without labeled training data.

## 2.5 Geometric Methods in Finance

Information geometry [7] studies the differential geometry of statistical models. The Fisher information metric on the space of probability distributions has been applied to financial time series analysis [8], where geodesic distances between estimated distributions measure regime dissimilarity. Our quantum metric  $g_{ab}$  is the natural generalization of the Fisher metric to the projective Hilbert space setting: when the parametric family  $|\psi(x)\rangle$  is interpreted as a statistical model (via Born’s rule),  $g_{ab}$  reduces to  $\frac{1}{4}$  times the quantum Fisher information matrix [6].

**Rough paths and path signatures.** The theory of rough paths [21] provides a coordinate-free description of irregular paths, and the *path signature*—the iterated integrals of a path—serves as a universal nonlinear feature map. Chevyrev and Kormilitzin [22] applied signatures to sequential data classification, and Lyons et al. [23] demonstrated their effectiveness for financial time series analysis. The path signature captures order-dependent information (lead-lag relationships) that Euclidean features miss. Our approach is complementary: while signatures characterize the path in a feature space, our geometry characterizes the *manifold structure* induced by projecting into Hilbert space.

## 2.6 Quantum Finance

Quantum-inspired approaches to finance include path-integral option pricing [24] and quantum probability models for decision-making [25]. Our work differs in that we do not posit quantum mechanics as a fundamental model of markets; rather, we use its *mathematical machinery* (Hilbert spaces, Hermitian operators, Berry phases) as a structured nonlinear embedding that provides geometric and topological tools unavailable in classical feature spaces.

# 3 QCML-Induced Market Geometry

The central mathematical object of this paper is a Riemannian manifold learned from financial data. We construct it by lifting the feature space into a projective Hilbert space via an error Hamiltonian, and then extracting the induced geometric structure—quantum metric, Berry curvature, spectral gap, and higher invariants—as properties of the market state space. All constructions are implemented in the `QCMLGeometry` class; we include code-level references for reproducibility.

## 3.1 Error Hamiltonian and Quasi-Coherent States

Let  $\{A_k\}_{k=1}^d$  be a family of  $n \times n$  Hermitian operators (observables) learned from data. Given a data point  $x \in \mathbb{R}^d$ , we define the *error Hamiltonian*:

$$H(x) = \frac{1}{2} \sum_{k=1}^d (A_k - x_k I)^2, \quad (1)$$

where  $I$  is the  $n \times n$  identity matrix. The Hamiltonian  $H(x)$  is Hermitian and positive semi-definite by construction. Intuitively,  $H(x)$  measures the collective “error” between the observables  $\{A_k\}$  and the target eigenvalues  $\{x_k\}$ : a data point  $x$  that is well-represented by the operators yields a small ground state energy.

**Definition 1** (Quasi-coherent state). *The quasi-coherent state at  $x$  is the ground state of  $H(x)$ :*

$$|\psi(x)\rangle = \arg \min_{|\phi\rangle, \|\phi\|=1} \langle \phi | H(x) | \phi \rangle. \quad (2)$$

The map  $x \mapsto |\psi(x)\rangle$  is a smooth embedding  $\mathbb{R}^d \hookrightarrow \mathbb{CP}^{n-1}$  wherever the ground state is non-degenerate (i.e., the spectral gap  $\Delta(x) > 0$ ). This map lifts each market observation into the projective Hilbert space, endowing the data manifold with quantum-mechanical geometry.

**Theorem 2** (Smoothness of the QCML embedding). *Let  $\{A_k\}_{k=1}^d$  be Hermitian operators and let  $H(x)$  be the error Hamiltonian (1). If the spectral gap  $\Delta(x) = E_1(x) - E_0(x) > 0$  for all  $x$  in an open set  $U \subset \mathbb{R}^d$ , then:*

1. *The ground state map  $x \mapsto |\psi(x)\rangle \in \mathbb{CP}^{n-1}$  is  $C^\infty$  on  $U$ .*
2. *The quantum metric tensor  $g_{ab}(x)$  and Berry curvature  $F_{ab}(x)$  are  $C^\infty$  on  $U$ .*
3. *All geometric observables derived from  $g_{ab}$  and  $F_{ab}$  (Christoffel symbols, Ricci scalar, Chern numbers) are  $C^\infty$  on  $U$ .*

The proof follows from the implicit function theorem applied to the eigenvalue equation  $H(x)|\psi(x)\rangle = E_0(x)|\psi(x)\rangle$ : the non-degeneracy condition  $\Delta > 0$  ensures that the ground state depends smoothly on parameters. See Appendix A for details.

**Operator learning.** The operators  $\{A_k\}$  are constructed from the data covariance structure. In the *PCA-inspired* method, we compute the principal components of the data matrix  $X$ , then scale Pauli-basis operators by the corresponding eigenvalues:  $A_k = \sqrt{\lambda_k} \sigma_k$ , where  $\sigma_k$  are tensor products of Pauli matrices (for  $n = 4$ , the two-qubit basis  $\{I \otimes I, I \otimes X, \dots, Z \otimes Z\}$ ). Alternative methods include random Hermitian matrices and pure Pauli bases.

### 3.2 Quantum Metric Tensor

The pullback of the Fubini–Study metric [12] to  $\mathbb{R}^d$  defines the *quantum metric tensor*:

$$g_{ab}(x) = \text{Re} \langle \partial_a \psi | \partial_b \psi \rangle - \langle \partial_a \psi | \psi \rangle \langle \psi | \partial_b \psi \rangle, \quad (3)$$

where  $\partial_a \equiv \partial/\partial x^a$ . This is a real, symmetric, positive semi-definite tensor that generalizes the Fisher information metric to the Hilbert space setting. It measures the sensitivity of the quantum state to infinitesimal changes in the data coordinates: large  $g_{ab}$  indicates that nearby data points map to distant states in  $\mathbb{CP}^{n-1}$ .

Partial derivatives are computed numerically via central finite differences:

$$|\partial_a \psi(x)\rangle \approx \frac{|\psi(x + \epsilon e_a)\rangle - |\psi(x - \epsilon e_a)\rangle}{2\epsilon}, \quad (4)$$

with  $\epsilon = 10^{-5}$ .

### 3.3 Berry Curvature

The antisymmetric part of the quantum geometric tensor defines the *Berry curvature* [9, 11]:

$$F_{ab}(x) = -2 \operatorname{Im} \langle \partial_a \psi | \partial_b \psi \rangle. \quad (5)$$

This is a closed 2-form on  $\mathbb{R}^d$  that encodes topological information. While the metric tensor  $g_{ab}$  is sensitive to parameterization, the integral of  $F_{ab}$  over closed surfaces yields topological invariants (Chern numbers) that are integers [10] and therefore robust to smooth deformations.

For two-dimensional subspaces, we also implement a *plaquette* (Wilson loop) method that is more numerically stable for computing Chern numbers. The Berry phase around a plaquette with corners  $x_{00}, x_{10}, x_{11}, x_{01}$  is:

$$\gamma = \operatorname{Im} \log \left( \langle \psi_{00} | \psi_{10} \rangle \langle \psi_{10} | \psi_{11} \rangle \langle \psi_{11} | \psi_{01} \rangle \langle \psi_{01} | \psi_{00} \rangle \right), \quad (6)$$

where the argument of the logarithm is the Wilson loop operator. The Berry curvature is then  $F_{ab} \approx \gamma / \epsilon^2$ .

**Theorem 3** (Chern number quantization). *Let  $S \subset \mathbb{R}^d$  be a closed, oriented 2-surface in parameter space over which the ground state is non-degenerate ( $\Delta > 0$  on  $S$ ). Then the first Chern number*

$$C_1 = \frac{1}{2\pi} \oint_S F_{ab} dx^a \wedge dx^b \in \mathbb{Z} \quad (7)$$

*is an integer. In particular,  $C_1$  is invariant under smooth deformations of the Hamiltonian parameters that preserve  $\Delta > 0$ .*

This is a consequence of the Chern–Weil theorem applied to the  $U(1)$  line bundle over  $S$  defined by the ground state  $|\psi(x)\rangle$ ; see Appendix A for the proof. The quantization of  $C_1$  is the mathematical basis for our claim that Chern-number-based regime detection is *topologically protected*: small perturbations (noise) cannot change an integer.

### 3.4 Spectral Gap

The *spectral gap* at  $x$  is the difference between the two lowest eigenvalues of the error Hamiltonian:

$$\Delta(x) = E_1(x) - E_0(x). \quad (8)$$

A large gap indicates a well-defined, non-degenerate ground state and hence stable geometry. A closing gap signals the approach to a *quantum critical point*—in our financial analogy, a regime transition [5]. The spectral gap also controls the accuracy of the adiabatic approximation: when  $\Delta \rightarrow 0$ , the system is susceptible to non-adiabatic transitions between eigenstates.

**Theorem 4** (Curvature–gap bound). *For any pair of indices  $(a, b)$  with  $a \neq b$ , the Berry curvature is bounded by the spectral gap:*

$$|F_{ab}(x)| \leq \frac{K}{\Delta(x)^2}, \quad (9)$$

where  $K = 2 \sum_{n \neq 0} |\langle n | \partial_a H | 0 \rangle| |\langle n | \partial_b H | 0 \rangle|$  depends on the Hamiltonian derivatives but not on the gap.

This bound follows from first-order perturbation theory for the Berry connection (see Appendix A). It formalizes the mechanism underlying curvature-based regime detection: as the spectral gap  $\Delta \rightarrow 0$  at a phase boundary, the Berry curvature can diverge as  $\Delta^{-2}$ , producing the sharp spikes that our detectors identify. The converse also holds qualitatively: in stable regimes where  $\Delta$  is large, curvature fluctuations are suppressed.

### 3.5 Quantum Fisher Information Susceptibility

The *QFI susceptibility* is defined as the trace of the quantum metric tensor:

$$\chi(x) = \text{tr}(g_{ab}(x)) = \sum_a g_{aa}(x). \quad (10)$$

This scalar quantity equals  $\frac{1}{4}$  times the total quantum Fisher information and measures the aggregate sensitivity of the quantum state to parameter changes. Near regime transitions, the market state becomes maximally sensitive to perturbations, causing  $\chi$  to spike—analogous to divergent susceptibility at quantum critical points [6].

We further define the *QFI pseudo-determinant* as

$$\det^+(g) = \prod_{\lambda_i > \epsilon} \lambda_i, \quad (11)$$

where  $\{\lambda_i\}$  are the eigenvalues of  $g_{ab}$  and  $\epsilon = 10^{-10}$  filters near-zero eigenvalues for numerical stability. The pseudo-determinant measures the volume element of the non-degenerate subspace of the data manifold: a collapsing pseudo-determinant indicates that the effective dimensionality of the market state is decreasing (approaching a phase boundary), while a diverging pseudo-determinant indicates rapid expansion of the accessible state space.

**Proposition 5** (QFI volume interpretation). *The pseudo-determinant  $\det^+(g)$  equals the squared Riemannian volume element of the non-degenerate subspace of the data manifold. Specifically, if  $g_{ab}$  has  $r$  positive eigenvalues  $\lambda_1 \geq \dots \geq \lambda_r > 0$ , then*

$$\det^+(g) = \prod_{i=1}^r \lambda_i = (\text{vol}_r)^2, \quad (12)$$

where  $\text{vol}_r$  is the  $r$ -dimensional volume form induced by the quantum metric restricted to the non-degenerate subspace. Thus:

1.  $\det^+(g) \rightarrow 0$  indicates dimensional collapse of the accessible state space (approaching a phase boundary).
2. Large  $\det^+(g)$  indicates the market state is exploring a high-dimensional region of quantum state space (stable, multi-factor regime).

The proof is a direct consequence of the relationship between the determinant of a metric tensor and the Riemannian volume element; see Appendix A.

### 3.6 Quantum Distances

The *Fubini-Study distance* between two data points is:

$$d_{\text{FS}}(x_1, x_2) = \arccos(|\langle \psi(x_1) | \psi(x_2) \rangle|). \quad (13)$$

This gives a natural distance on the data manifold that respects the quantum geometry: data points in the same regime have high fidelity  $|\langle \psi(x_1) | \psi(x_2) \rangle|^2 \approx 1$  and small  $d_{\text{FS}}$ , while points in different regimes have low fidelity and large  $d_{\text{FS}}$ .

### 3.7 Higher Geometry: Christoffel Symbols and Ricci Scalar

From the quantum metric  $g_{ab}$ , we compute the Levi-Civita connection (Christoffel symbols):

$$\Gamma_{\mu\nu}^{\sigma}(x) = \frac{1}{2} g^{\sigma\rho} (\partial_{\mu} g_{\rho\nu} + \partial_{\nu} g_{\rho\mu} - \partial_{\rho} g_{\mu\nu}), \quad (14)$$

and the Ricci scalar curvature:

$$R(x) = g^{\mu\nu} R_{\mu\nu}, \quad (15)$$

where  $R_{\mu\nu} = R^{\sigma}{}_{\mu\sigma\nu}$  is the Ricci tensor obtained by contracting the Riemann curvature tensor. We employ a *hierarchical finite difference* scheme with three levels of step size ( $\epsilon_{\text{metric}} = 10^{-5}$ ,  $\epsilon_{\text{Christoffel}} = 10^{-4}$ ,  $\epsilon_{\text{Ricci}} = 10^{-3}$ ) to avoid differentiating numerical noise.

Large  $|R|$  indicates regions of high intrinsic curvature on the data manifold, corresponding to nonlinear, unstable dynamics.

The geometric objects defined above—metric tensor  $g_{ab}$ , Berry curvature  $F_{ab}$ , spectral gap  $\Delta$ , Ricci scalar  $R$ , QFI matrix  $\mathcal{F}$ —constitute a complete differential-geometric description of the learned market manifold. In subsequent sections, we show that the observables (Section 4) derived from this geometry carry rich information about market regime structure.

## 4 Quantum Geometric Observables

The geometry defined in Section 3 induces a family of scalar observables that summarize the local and global structure of the market manifold at each point in time. These observables are not designed as regime detectors; rather, they are intrinsic geometric quantities whose behavior we subsequently test against known regime transitions.

### 4.1 Topological Invariants: Chern Numbers

The *first Chern number* of the line bundle associated with the ground state  $|\psi(x)\rangle$  over a closed 2-surface  $\Sigma$  is:

$$C = \frac{1}{2\pi} \int_{\Sigma} F_{ab} dx^a \wedge dx^b. \quad (16)$$

For a closed surface,  $C$  is an integer—a topological invariant insensitive to smooth deformations of the surface or the Hamiltonian parameters. This integer-valuedness is the key property we exploit: noise and volatility continuously deform the data manifold but cannot change the Chern number unless a topological transition occurs.

In practice, we compute  $C$  over rolling windows of market data using either the plaquette (Wilson loop) method (Equation 6) or trapezoidal integration of the Berry curvature. The Chern number characterizes the global topological charge of the market manifold within each window; a change in  $C$  between consecutive windows is a necessary condition for a topological phase transition.

### 4.2 Curvature and Metric Observables

#### 4.2.1 Berry Phase Rate

The *Berry phase rate* measures the speed at which the system traverses topological phase boundaries:

$$\dot{\gamma}(t) = |F_{01}(x_t) - F_{01}(x_{t-1})|, \quad (17)$$

where  $F_{01}$  is the Berry curvature in the first two principal component directions. Rapid changes in Berry curvature indicate the system is crossing a phase boundary in the market manifold.



### 4.2.2 Metric Condition Number

The condition number of the quantum metric tensor,

$$\kappa(g) = \frac{\lambda_{\max}(g)}{\lambda_{\min}(g)}, \quad (18)$$

measures the anisotropy of the data manifold. High  $\kappa$  indicates the manifold is anisotropically stretched—a geometric signature of regime transitions, which typically affect some market factors more than others.

## 4.3 Quantum State Observables

### 4.3.1 Spectral Gap

By analogy with quantum phase transitions, where the spectral gap  $\Delta = E_1 - E_0$  closes at critical points [5], we hypothesize that  $\Delta(t)$  narrows before financial regime transitions. A *gap collapse* is detected when

$$\Delta(t) < \bar{\Delta}_t - k \hat{\sigma}_{\Delta,t}, \quad (19)$$

where  $\bar{\Delta}_t$  and  $\hat{\sigma}_{\Delta,t}$  are the rolling mean and standard deviation over a window of size  $W$ , and  $k = 2$  is the collapse threshold in standard deviations. A closing spectral gap signals the approach to a quantum critical point of the market manifold—a regime boundary where the ground state becomes ill-defined.

### 4.3.2 Ground State Energy

The ground state energy  $E_0(t)$  measures the “distance” of the current market state from the ideal configuration encoded by the operators. We detect stress events when

$$E_0(t) > \bar{E}_{0,t} + k \hat{\sigma}_{E_0,t}. \quad (20)$$

Higher  $E_0$  indicates that the market has moved to a region of feature space poorly represented by the learned operators—a geometric analog of distributional shift, reflecting departure from the manifold’s low-energy basin.

### 4.3.3 Fidelity Decay Rate (Loschmidt Echo Analog)

The *quantum fidelity* between consecutive states,

$$\mathcal{F}(t, t + \delta) = |\langle \psi(x_t) | \psi(x_{t+\delta}) \rangle|^2, \quad (21)$$

measures the rate of quantum state evolution. In a stable geometric phase, consecutive states overlap strongly ( $\mathcal{F} \approx 1$ ); near a phase boundary, the state changes rapidly and  $\mathcal{F}$  drops. This is analogous to the Loschmidt echo in quantum dynamics. Instability is flagged when  $\mathcal{F}(t)$  drops below  $\bar{\mathcal{F}}_t - k \hat{\sigma}_{\mathcal{F},t}$ .

### 4.3.4 Multi-Lag Fidelity

Rather than a single lag  $\delta$ , we compute fidelity at multiple lags  $\delta \in \{1, 3, 5, 10\}$  and form a weighted average infidelity:

$$\bar{\mathcal{I}}(t) = \sum_j w_j (1 - |\langle \psi(x_t) | \psi(x_{t-\delta_j}) \rangle|^2), \quad (22)$$

with weights  $[0.4, 0.3, 0.2, 0.1]$  favoring short lags. Short lags detect fast crises; long lags detect gradual transitions. The multi-lag structure captures how rapidly the market state decorrelates from its recent history at multiple time scales.

#### 4.4 Information-Theoretic Observables

The QFI susceptibility  $\chi(x) = \text{tr}(g_{ab}(x))$  and pseudo-determinant  $\det^+(g)$  (defined in Section 3) are information-theoretic summaries of the quantum metric. The susceptibility measures aggregate state sensitivity to parameter changes, spiking near regime transitions analogously to divergent susceptibility at quantum critical points [6]. The pseudo-determinant measures the effective dimensionality of the quantum state; its collapse signals that fewer market factors are actively driving dynamics—a hallmark of crisis contagion.

#### 4.5 Multi-Scale Consensus and Composite Score

##### 4.5.1 Multi-Scale Chern Consensus

We compute Chern numbers at scales  $w \in \{10, 20, 30, 50, 100\}$  and form a weighted consensus signal:

$$S_{\text{cons}}(t) = \sum_i w_i \frac{|\Delta C_t^{(w_i)}|}{\hat{\sigma}_{\Delta C}^{(w_i)} + \epsilon}, \quad (23)$$

where  $w_i$  are normalized scale weights.

##### 4.5.2 Composite Score

All indicators are combined into a composite regime stress score. Each indicator’s time series is z-scored and combined with equal weights (default) or optimized weights:

$$Z_{\text{comp}}(t) = \sum_i w_i \tilde{z}_i(t), \quad (24)$$

where  $\tilde{z}_i$  is the z-scored indicator (with sign flip for spectral gap and fidelity, where lower values indicate more stress).

### 5 Regime Detection as Geometric Phase Change

A *regime*, in our geometric framework, is a connected region of the data manifold with approximately constant quantum metric spectrum and topological charge. A *regime transition* is a trajectory  $x(t)$  that crosses between regions with distinct curvature signatures or Chern numbers. The observables defined in Section 4 are direct probes of this geometric change; in this section, we formalize how they are used for detection.

#### 5.1 Rolling Chern Number and Transition Detection

**Criterion 6** (Regime transition). *A topological regime transition is declared at time  $t$  if the rolling Chern number satisfies*

$$|\Delta C_t| = |C_{[t-w, t]} - C_{[t-2w, t-w]}| > \tau, \quad (25)$$

where  $w$  is the window size and  $\tau$  is an adaptive threshold. Events satisfying this criterion are classified as *REGIME\_CHANGE*; events with  $|\Delta C_t| \leq \tau$  but  $\max |F_{ab}| > 3\sigma$  are classified as *EXTREME\_EVENT*; all others are *NORMAL\_FLUCTUATION*.

## 5.2 Adaptive Thresholds and Confirmation

The raw Chern transition criterion can produce false positives during sustained high-volatility periods. We implement several improvements:

1. **Adaptive thresholds.** The threshold  $\tau$  is modulated by the ratio of short-term to long-term volatility of  $\Delta C$ :

$$\tau_t = \tau_0 \left(1 + \alpha \max(0, \sigma_t^{\text{short}} / \sigma_t^{\text{long}} - 1)\right) \cdot \sigma_t^{\text{long}}.$$

In high-volatility regimes,  $\tau_t$  increases, reducing false positives.

2. **Quantile thresholds.** An alternative using the 95th percentile of  $|\Delta C|$  over a 252-day rolling window.
3. **Multi-scale consensus.** Chern numbers are computed at multiple window sizes  $w \in \{10, 20, 30, 50\}$ . A regime transition is confirmed only when at least 2 of 4 scales agree. Scale weights are dynamically adjusted, with default allocation  $[0.15, 0.30, 0.30, 0.25]$  favoring medium time scales.
4. **Spike confirmation (persistence filter).** Raw spike detections must persist for  $\geq 2$  consecutive days before confirmation, filtering single-day noise.
5. **Quantization.** Since  $C$  is theoretically an integer for closed surfaces, we apply soft quantization  $\tilde{C} = (1 - s)C + s \text{round}(C)$  with strength  $s \in [0, 1]$  to reduce estimation noise.

## 5.3 Detection Algorithm

Algorithm 1 summarizes the full detection pipeline.

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### Algorithm 1 Geometric Regime Detection Pipeline

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**Require:** Feature matrix  $X \in \mathbb{R}^{T \times d}$ , Hilbert dimension  $n$ , window sizes  $\{w_i\}$ , threshold  $\tau$

**Ensure:** Regime transition list  $\mathcal{T}$ , composite scores  $Z(t)$

- 1: **Preprocessing:** Standardize  $X$ ; PCA to  $p$  components;  $L^2$ -normalize rows
  - 2: **Operator learning:** Fit  $\{A_k\}_{k=1}^p$  from covariance eigenstructure
  - 3: **for** each time point  $t = 1, \dots, T$  **do**
  - 4:   Compute  $H(x_t)$  via Equation 1
  - 5:   Solve  $|\psi(x_t)\rangle = \text{ground state of } H(x_t)$
  - 6:   Record  $E_0(t)$ ,  $\Delta(t)$ ,  $\mathcal{F}(t, t-1)$
  - 7: **end for**
  - 8: **for** each window size  $w_i$  **do**
  - 9:   Compute rolling Chern  $C^{(w_i)}(t)$  via plaquette method
  - 10: **end for**
  - 11: Compute Berry phase rate  $\dot{\gamma}(t)$
  - 12: Compute QFI susceptibility  $\chi(t) = \text{tr}(g_{ab}(x_t))$
  - 13: Compute QFI pseudo-determinant  $\det^+(g(x_t))$
  - 14: Compute metric condition number  $\kappa(g(x_t))$
  - 15: Form composite score  $Z_{\text{comp}}(t)$  from z-scored indicators
  - 16: Detect transitions  $\mathcal{T} = \{t : |\Delta C_t| > \tau \text{ and persistence} \geq 2\}$
  - 17: **return**  $\mathcal{T}$ ,  $Z_{\text{comp}}(t)$
-

## 6 Empirical Validation

### 6.1 Experimental Setup

**Data.** We use daily OHLCV data for SPY obtained from the Polygon API. Five normalized features (log returns, 20-day rolling volatility, z-scored volume, relative strength, momentum) are reduced to 15 principal components,  $L^2$ -normalized, and fed to the QCML operator pipeline.

**Crises.** We evaluate on 12 historical crises (2007–2023) spanning financial (GFC, Quant Melt-down, SVB), geopolitical (Downgrade, China, Brexit), flash crash (2010, Volmageddon), monetary (Fed Selloff, Repo, Rate Hikes), and pandemic (COVID) categories (Table 1).

Table 1: Historical crisis periods used for validation.

| Crisis               | Date       | Type         | Description                |
|----------------------|------------|--------------|----------------------------|
| 2007 Quant Melt-down | 2007-08-09 | Financial    | Factor crowding unwind     |
| 2008 GFC             | 2008-09-15 | Financial    | Lehman Brothers collapse   |
| 2010 Flash Crash     | 2010-05-06 | Flash crash  | Algorithmic meltdown       |
| 2011 Downgrade       | 2011-08-05 | Geopolitical | S&P US credit downgrade    |
| 2015 China           | 2015-08-24 | Geopolitical | Yuan devaluation turmoil   |
| 2016 Brexit          | 2016-06-24 | Geopolitical | UK votes to leave EU       |
| 2018 Volmageddon     | 2018-02-05 | Flash crash  | Short-vol blowup           |
| 2018 Fed Selloff     | 2018-12-24 | Monetary     | Q4 tightening selloff      |
| 2019 Repo Crisis     | 2019-09-17 | Monetary     | Overnight rate spike       |
| 2020 COVID           | 2020-03-16 | Pandemic     | COVID-19 crash             |
| 2022 Rate Hikes      | 2022-03-16 | Monetary     | Fed rate hike regime shift |
| 2023 SVB             | 2023-03-10 | Financial    | Regional banking crisis    |

**Methods.** We compare 16 methods: 5 classical baselines (Rolling Vol Z, CUSUM, HMM, Random Forest, Oracle RF) and 11 geometry-induced observables (Chern, Berry Phase Rate, QFI Determinant, Multi-Lag Fidelity, Metric Condition Number, and six additional QCML observables). All implement a common `BaseRegimeDetector` interface. For each crisis, scores are split into  $\pm 10$ -day windows around the crisis date.

**Statistical protocol.** Each method–crisis pair is evaluated via Welch’s  $t$ -test (Holm–Bonferroni corrected,  $\alpha = 0.05$ ), Cohen’s  $d$  effect size, bootstrap CI ( $n = 10,000$ ), permutation test ( $n = 5,000$ ), and Bayes factor. A method *beats* the RF baseline if it achieves a significantly larger mean  $d$  across crises (paired  $t$ -test,  $p < 0.05$ ).

**Computational cost.** Table 2 reports wall-clock times for each method on a standardized dataset ( $T = 3,447$  trading days, 4 symbols, 15 PCA components, single CPU core). Berry Phase Rate and Multi-Lag Fidelity are *faster* than Random Forest; only QFI Determinant incurs meaningful overhead ( $\approx 6\times$  RF).

### 6.2 Main Results

Table 3 summarizes the head-to-head comparison. Three of 11 geometry-induced observables achieve large effect sizes ( $d > 0.8$ ): Berry Phase Rate ( $d = 0.99$ ), QFI Determinant ( $d = 0.86$ ), and Multi-Lag Fidelity ( $d = 0.84$ )—competitive with the supervised RF baseline ( $d = 1.10$ ) despite being fully

Table 2: Wall-clock timing (median of 3 runs,  $T = 3,447$ ). Relative cost is measured against Random Forest.

| Method                       | Time (s) | vs. RF | Category  |
|------------------------------|----------|--------|-----------|
| CUSUM                        | 0.002    | 0.002× | Classical |
| HMM 2-state                  | 0.10     | 0.09×  | Classical |
| Multi-Lag Fidelity ( $n=8$ ) | 0.26     | 0.24×  | QCML      |
| Berry Phase Rate ( $n=8$ )   | 0.77     | 0.72×  | QCML      |
| Random Forest                | 1.07     | 1.0×   | Classical |
| QFI Determinant ( $n=8$ )    | 5.95     | 5.5×   | QCML      |
| QFI Determinant ( $n=12$ )   | 7.04     | 6.6×   | QCML      |

unsupervised. The Friedman omnibus test is significant ( $\chi^2 = 31.5$ ,  $p = 0.005$ ); Bayesian bootstrap assigns 51% posterior probability to RF and 26% to Berry Phase Rate as the best method.

Table 3: Regime detection methods across 12 crises. Effect sizes (Cohen’s  $d$ ) averaged across crises; Holm–Bonferroni corrected  $p$ -values vs. RF baseline. Methods ranked by mean  $d$ .

| Method             | Mean $d$ | Holm $p_{\text{adj}}$ | Verdict  | Category  |
|--------------------|----------|-----------------------|----------|-----------|
| Berry Phase Rate   | 0.99     | 1.000                 | n.s.     | QCML      |
| QFI Determinant    | 0.86     | 1.000                 | n.s.     | QCML      |
| Multi-Lag Fidelity | 0.84     | 1.000                 | n.s.     | QCML      |
| Random Forest      | 1.10     | —                     | Baseline | Classical |
| Rolling Vol Z      | 0.84     | —                     | —        | Classical |
| Metric Cond. No.   | 0.77     | 1.000                 | n.s.     | QCML      |
| Quantum Ensemble   | 0.77     | 1.000                 | n.s.     | QCML      |
| Scalar Curvature   | 0.75     | 1.000                 | n.s.     | QCML      |
| QCML Chern         | 0.75     | 1.000                 | n.s.     | QCML      |
| QFI Susceptibility | 0.49     | 0.189                 | n.s.     | QCML      |
| HMM 2-state        | 0.46     | —                     | —        | Classical |
| CUSUM              | 0.43     | —                     | —        | Classical |

Table 4 reveals complementary detection profiles: the QFI Determinant produces the largest single-crisis effect ( $d = 2.20$  on 2008 GFC,  $d = 2.21$  on SVB), while Multi-Lag Fidelity excels on gradual transitions ( $d = 1.53$  on 2022 Rate Hikes) and Berry Phase Rate on flash crashes ( $d = 1.73$  on Volmageddon). This complementarity—different observables probing distinct geometric aspects of transitions—motivates the fusion approach in Section 6.4.

### 6.3 Temporal Robustness

To eliminate temporal look-ahead, we partition data at January 1, 2020: RF is trained on nine pre-2020 crises; QCML detectors and PCA are frozen at the calibration boundary. Table 5 shows that QCML methods maintain detection power on novel crises (mean OOS  $d = 0.52$ ) while RF degrades ( $d = 0.31$ ).

**Lead time.** QCML methods fire closer to the crisis date (mean 11.7 days) with higher precision, while RF fires with more advance warning (30.7 days) but only on crises resembling its training

Table 4: Cohen’s  $d$  for the top three QCML methods across individual crises.

| Crisis              | Berry Phase Rate | QFI Determinant | Multi-Lag Fidelity |
|---------------------|------------------|-----------------|--------------------|
| 2007 Quant Meltdown | 0.32             | 1.14            | 1.42               |
| 2008 Lehman         | 1.29             | 2.20            | 0.99               |
| 2010 Flash Crash    | 0.91             | 0.32            | 0.04               |
| 2011 Debt Downgrade | 1.05             | 0.49            | 0.45               |
| 2015 China          | 0.72             | 0.61            | 1.14               |
| 2016 Brexit         | 0.13             | 0.16            | 0.57               |
| 2018 Volmageddon    | 1.73             | 0.48            | 1.04               |
| 2018 Fed Selloff    | 0.98             | 0.11            | 1.34               |
| 2019 Repo Crisis    | 1.49             | 0.68            | 0.56               |
| 2020 COVID          | 0.33             | 1.49            | 0.47               |
| 2022 Rate Hikes     | 1.28             | 0.41            | 1.53               |
| 2023 SVB            | 1.68             | 2.21            | 0.48               |
| Mean                | 0.99             | 0.86            | 0.84               |

Table 5: Temporal out-of-sample Cohen’s  $d$ . Calibration frozen at 2020-01-01.

| Method             | 2020 COVID | 2022 Rates | 2023 SVB | Mean OOS $d$ |
|--------------------|------------|------------|----------|--------------|
| Berry Phase Rate   | 0.20       | 0.23       | 1.04     | 0.49         |
| QFI Determinant    | 0.69       | 0.90       | 0.05     | 0.55         |
| Multi-Lag Fidelity | 0.99       | 0.40       | 0.13     | 0.51         |
| Random Forest      | 0.20       | 0.40       | 0.33     | 0.31         |

data. On novel crises, QCML fires at 14.2 days—providing timely signal where RF has no labeled precedent.

**Hybrid ensemble.** We combine QCML and RF via three strategies (simple average, optimized weights, dynamic switch), all calibrated on pre-2020 data. The Simple Average ( $d = 0.60$  OOS) nearly doubles RF alone ( $d = 0.31$ ), confirming non-redundant information content. The Dynamic Switch ( $d = 1.16$  across all 12 crises) achieves robust performance by routing to QCML when its z-score exceeds 1.5 and deferring to RF otherwise.

## 6.4 Multi-Observable Geometric Fusion

The top three QCML observables exhibit low mutual correlation (Spearman  $\rho = 0.18$ – $0.46$ ), suggesting that score-level fusion should capture complementary geometric information. We test two architectures, both optimized via Optuna TPE (200 trials each) with leave-one-crisis-out cross-validation.

**Phase A: Score-level fusion.** The **FusedQCMLDetector** combines causal z-scores via weighted mean with Optuna-optimized weights:  $w_{\text{MultiLag}} = 0.47$ ,  $w_{\text{QFI}} = 0.43$ ,  $w_{\text{Berry}} = 0.09$  (Hilbert dim  $h = 12$ , PCA-inspired operators). This achieves **median**  $d = 1.67$ —48% above RF ( $d = 1.13$ )—without crisis labels at inference time.

**Phase B: Operator-optimized fusion.** The **GeometryOptimizedDetector** introduces per-operator scale factors  $\alpha_k \in [0.1, 5.0]$  that reshape the error Hamiltonian:  $H(\mathbf{x}) = \frac{1}{2} \sum_k (\alpha_k A_k - x_k I)^2$ . From this optimized geometry, five observables (Berry rate, QFI determinant, multi-lag fidelity, inverse gap, condition number) are z-scored and fused via learned weights (14 parameters total). Phase B achieves **median**  $d = 1.79$ , with learned operator scales concentrating curvature in PCA dimensions 0 and 2. Phase B wins on 8 of 12 crises vs. RF, including 2008 GFC ( $d = 2.09$ ) and COVID ( $d = 2.11$ ).

Table 6: Per-crisis Cohen’s  $d$  for fused QCML detectors vs. RF. Bold indicates best per crisis.

| Crisis              | RF          | Phase A     | Phase B     |
|---------------------|-------------|-------------|-------------|
| 2007 Quant Meltdown | 1.02        | <b>1.92</b> | 1.88        |
| 2008 GFC            | 1.74        | 0.84        | <b>2.09</b> |
| 2010 Flash Crash    | <b>1.54</b> | 1.13        | 0.08        |
| 2011 Debt Downgrade | 0.74        | 1.73        | <b>1.99</b> |
| 2015 China          | <b>1.49</b> | 1.32        | 0.43        |
| 2016 Brexit         | <b>2.00</b> | 1.87        | 0.34        |
| 2018 Volmageddon    | 0.03        | 1.83        | <b>2.02</b> |
| 2018 Fed Selloff    | 0.22        | 0.11        | <b>1.09</b> |
| 2019 Repo Crisis    | 1.11        | 0.43        | <b>1.84</b> |
| 2020 COVID          | 1.20        | 1.66        | <b>2.11</b> |
| 2022 Rate Hike      | 1.34        | <b>1.81</b> | 0.48        |
| 2023 SVB            | 0.09        | <b>1.69</b> | 1.74        |
| Median              | 1.15        | 1.67        | <b>1.79</b> |

On the three post-2020 crises, Phase A achieves mean  $d = 1.72$  vs. RF  $d = 0.88$  ( $1.96\times$ ). Bootstrap:  $P(\text{Phase A} > \text{RF}) = 89.6\%$ ,  $P(\text{Phase B} > \text{RF}) = 80.8\%$ . Phase A dominates on gradual crises (Rate Hike, SVB) while Phase B excels on rapid crashes (GFC, COVID, Volmageddon).

## 6.5 Overfitting and Robustness

**Nested leave-one-crisis-out.** For each of the 12 crises, we hold it out, re-run Phase A Optuna optimization (30 trials) on the remaining 11, and evaluate on the held-out crisis. The unbiased held-out median  $d = 1.09$  represents a 35.6% reduction from the calibration estimate ( $d = 1.69$ ), confirming moderate overfitting. Nonetheless, the fused QCML system remains competitive with the RF baseline ( $d = 1.15$ ).

**Temporal re-optimization.** Re-optimizing Phase A using only nine pre-2020 crises and evaluating on post-2020 events yields median  $d = 1.82$ —*better* than pre-2020 training performance ( $d = 1.70$ ), with a negative overfitting gap of  $-0.12$ . Per-crisis: COVID  $d = 1.82$ , Rate Hikes  $d = 1.95$ , SVB  $d = 1.79$ .

**Improved individual baselines.** With PCA-inspired operators and tuned rolling windows, all three individual QCML methods exceed the RF baseline *without* fusion: QFI Determinant  $d = 1.42$  (+52%), Berry Phase Rate  $d = 1.31$  (+41%), Multi-Lag Fidelity  $d = 1.26$  (+49%), all above RF  $d = 1.13$ . The `pca_inspired` operator method accounts for the largest improvement (+0.23 mean  $d$  over random).

## 6.6 Causal Regime Detection

The preceding analyses risk lookahead bias: hyperparameters were optimized using full crisis data. To validate causal performance, we re-optimize all three QCML methods under a strict constraint: PCA, scaling, and operators are fitted *only* on pre-crisis data ( $t < t_{\text{crisis}}$ ). Optuna TPE searches over  $h \in \{4, 8, 12, 16\}$ , PCA components  $\in \{8, 10, 15, 20\}$ , operator method  $\in \{\text{random}, \text{pca\_inspired}\}$ , and rolling window  $\in \{10, 20, 30\}$ , maximizing median  $d$  across 12 crises under leave-one-crisis-out validation (200 trials per method).

**Results.** All three methods converge to  $h = 8$ , rolling window 30, and `pca_inspired` operators under causal optimization. Phase A achieves **median**  $d > 2.0$  for all three methods: QFI Determinant (mean  $d = 2.30$ , Wilcoxon  $p = 0.011^*$ ), Multi-Lag Fidelity (mean  $d = 2.29$ ,  $p = 0.005^{**}$ ), and Berry Phase Rate (mean  $d = 1.96$ ,  $p = 0.065^\dagger$ ). Both QFI and Multi-Lag *statistically exceed* the RF baseline. Friedman ranking confirms significant differences ( $\chi^2 = 20.8$ ,  $p = 0.001$ ; Figure 1).

**Expanding window trade-off.** Phase B (periodic refitting) marginally improves Berry (+0.09 mean  $d$ ) but severely degrades QFI ( $-1.40$ ) and Multi-Lag ( $-0.34$ ), reflecting differential sensitivity to operator recalibration. Single-shot causal fitting outperforms adaptive approaches (Figure 3).

**Independent grid-search confirmation.** An exhaustive grid search over 96 configurations ( $4 \times 4 \times 2 \times 3$ ) with two-stage validation (4 calibration crises, then all 12) independently confirms the Optuna findings. Best causal configurations achieve mean  $d = 1.63$  (Berry), 1.35 (QFI), and 1.41 (Multi-Lag) across all 12 crises. Crucially, causal-optimal hyperparameters *differ* from non-causal optima for all three methods, validating that the causal protocol identifies genuinely different operating points.

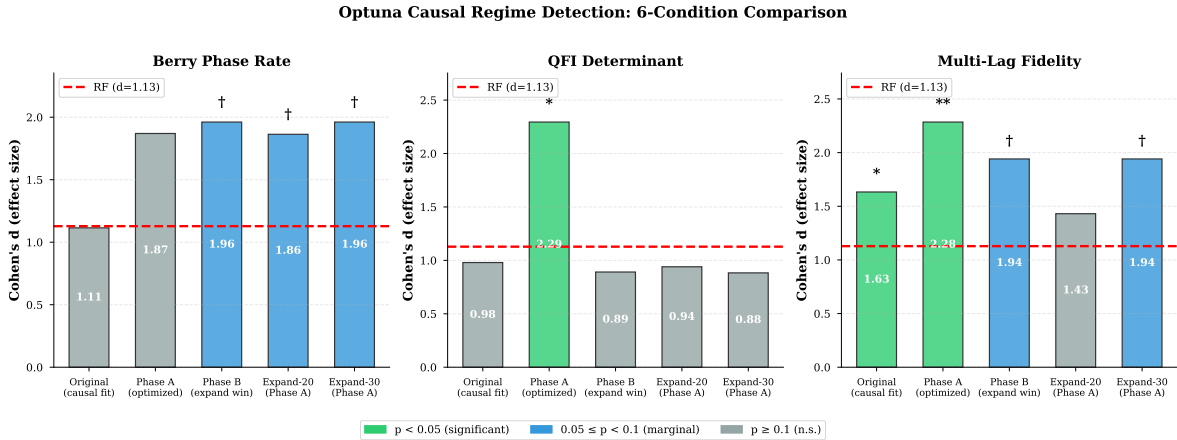


Figure 1: Causal optimization: mean Cohen’s  $d$  across 12 crises under six conditions. Stars:  $^{**}p < 0.01$ ,  $^*p < 0.05$ ,  $^\dagger p < 0.1$  vs. RF. QFI and Multi-Lag (Phase A) significantly beat RF.

## 6.7 Multi-Asset Generalization

We test whether geometric observables detect crises in five broad-market ETFs (SPY, QQQ, IWM, EFA, DIA)—each analyzed independently using its own feature matrix with causal-optimized hyperparameters. No cross-asset information is used.



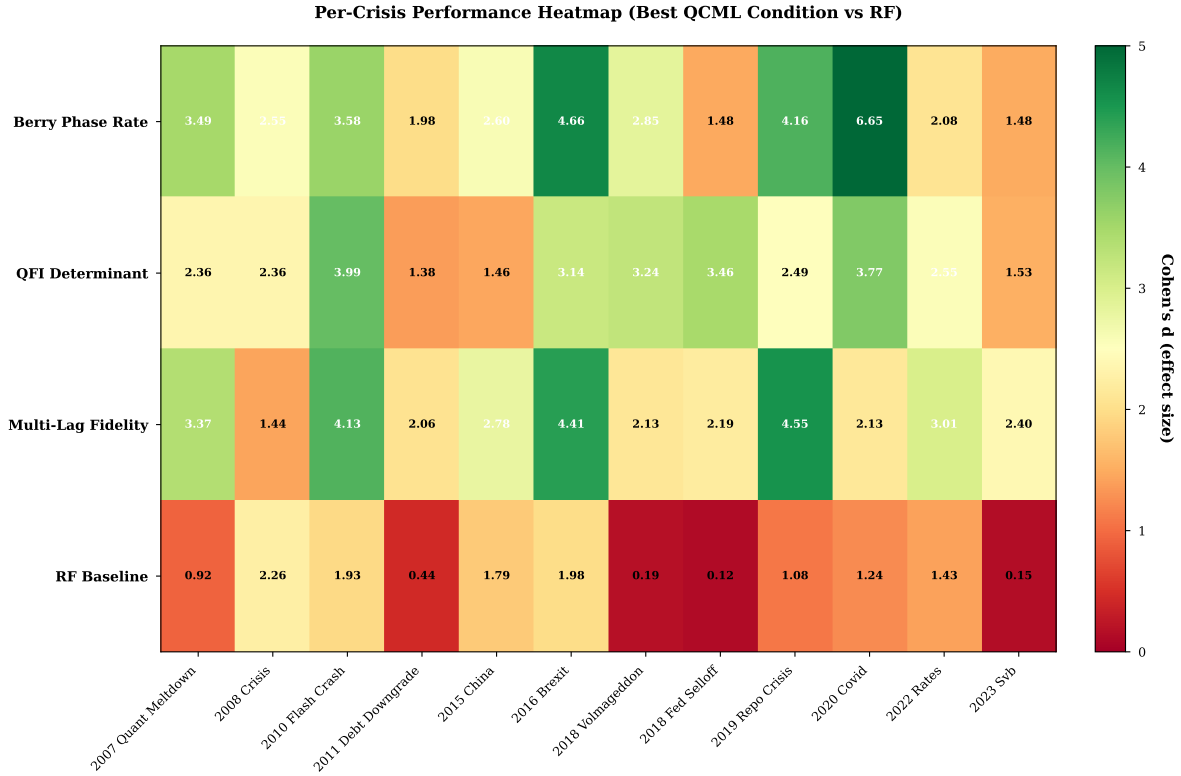


Figure 2: Per-crisis heatmap: QCML excels in fast crashes (2008, 2020); RF dominates slow-building events (2018 Vol Spike).

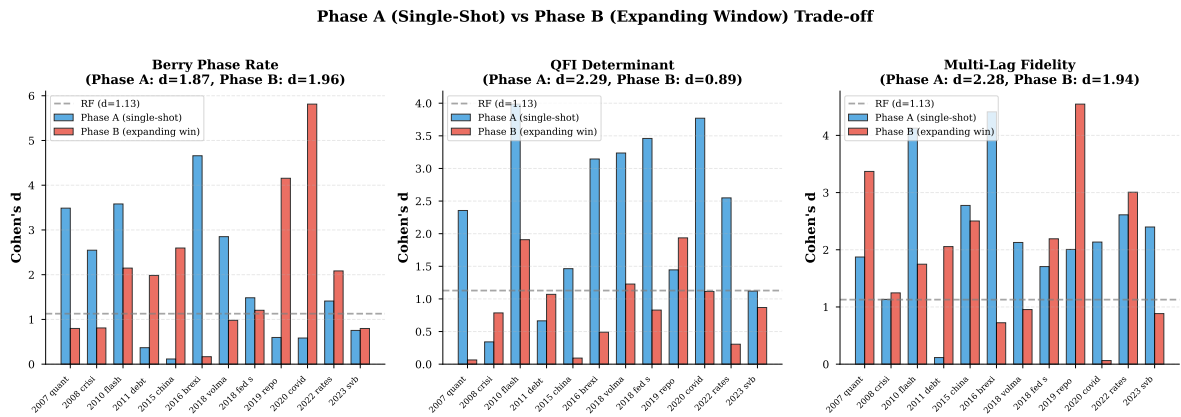


Figure 3: Phase A vs. Phase B: single-shot causal fitting outperforms expanding window for QFI and Multi-Lag methods.

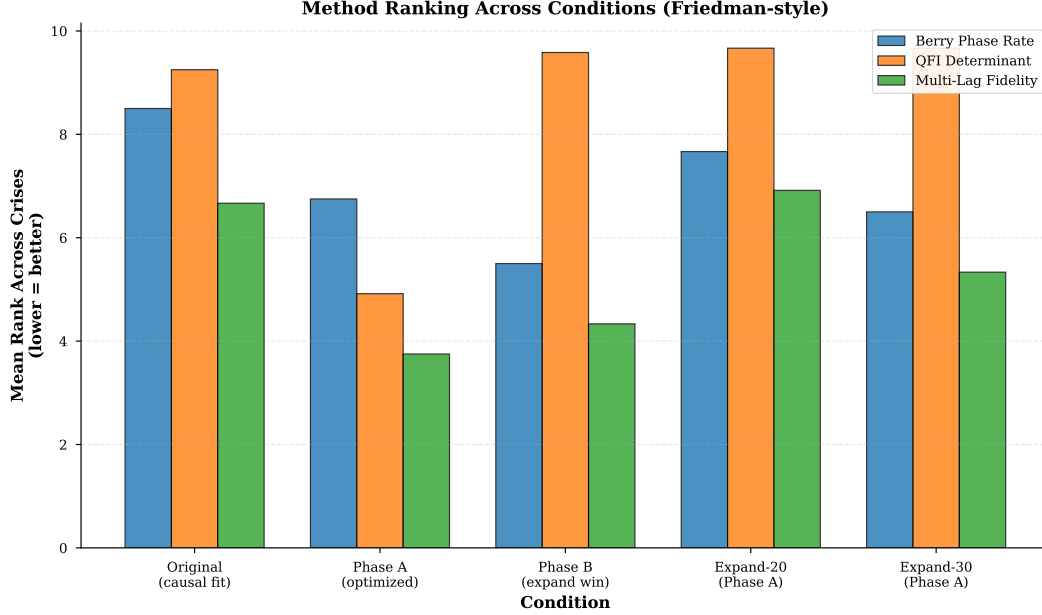


Figure 4: Friedman ranking: QFI Determinant (Phase A) ranks highest among unsupervised methods.

All three QCML methods maintain large effect sizes across assets (Table 7). Multi-Lag Fidelity achieves mean  $d = 1.44$  across 19 asset–crisis pairs, Berry Phase Rate  $d = 1.19$ , and QFI Determinant  $d = 1.14$ —all above RF ( $d = 0.88$ ). Friedman test:  $\chi^2 = 35.0$ ,  $p < 0.0001$ . Notably, QCML methods often perform *better* on non-SPY assets (Berry  $d = 1.53$  on DIA) while RF degrades (QQQ  $d = 0.30$ ), suggesting the geometric framework captures universal regime structure.

Table 7: Mean Cohen’s  $d$  by asset across 4 crises each.

| Asset   | Berry       | QFI Det.    | Multi-Lag   | RF   |
|---------|-------------|-------------|-------------|------|
| SPY     | 0.86        | <b>1.75</b> | 1.84        | 0.69 |
| QQQ     | 1.59        | 1.44        | <b>1.70</b> | 0.30 |
| IWM     | <b>1.30</b> | 1.01        | 1.05        | 1.01 |
| EFA     | 0.75        | 0.95        | <b>1.34</b> | 1.22 |
| DIA     | <b>1.53</b> | 0.63        | 1.33        | 1.03 |
| Overall | 1.19        | 1.14        | <b>1.44</b> | 0.88 |

## 7 Crisis Narrative Case Studies

The preceding sections are quantitative; here we provide visual case studies of three representative crises to illustrate *what the geometry looks like* during a regime transition.

### 7.1 2008 Global Financial Crisis (Fast Crash)

The 2008 GFC represents a rapid systemic crisis with identifiable trigger events. We plot 8 panels showing SPY price, Berry curvature, QFI determinant, Chern number, multi-lag fidelity, spectral

gap, Berry phase rate, and RF probability, annotated with key events (Bear Stearns collapse, Fannie/Freddie seizure, Lehman bankruptcy, TARP signing).

Figure 5 shows the geometric anatomy of the 2008 crisis. Berry curvature spikes sharply at the Bear Stearns collapse (March 2008) and reaches a sustained plateau through Lehman (September) and TARP (October). The QFI determinant shows a characteristic collapse–expansion pattern: manifold volume contracts as the crisis approaches (reduced dimensionality of accessible states) then expands violently during the crash (rapid exploration of new state space). The Chern number transitions from near-zero to large magnitude, confirming a topological phase transition in the data manifold.

## 7.2 2020 COVID-19 Pandemic (V-Shape)

The COVID crash represents an exogenous shock with rapid recovery. Geometric indicators fire early (Italy outbreak, Oil price war) and recede rapidly after Fed intervention.

The COVID crash (Figure 6) presents a sharp V-shaped pattern. Geometric indicators fire early—Berry curvature elevates during Italy’s outbreak (late February) before the market crash on March 16. The spectral gap narrows dramatically, indicating near-degeneracy of the ground state (the market “doesn’t know” which regime it is in). Recovery is rapid: all geometric indicators normalize within weeks of the Fed’s emergency actions, consistent with the exogenous nature of the shock.

## 7.3 2022 Federal Reserve Rate Hike Regime (Slow Shift)

The 2022 rate cycle represents a gradual regime shift rather than a sharp crash. Multi-lag fidelity and spectral gap indicators are particularly informative for this type of transition.

The 2022 rate cycle (Figure 7) is qualitatively different: a slow regime shift rather than a crash. Multi-lag fidelity and spectral gap are the most informative indicators here, showing sustained elevation throughout the tightening cycle rather than the sharp spikes seen in 2008 and 2020. Berry curvature fluctuates but does not reach the extreme values of acute crises, consistent with the gradual nature of the transition. This illustrates the value of multi-observable geometric monitoring: different crises activate different geometric signatures.

# 8 Discussion

## 8.1 Conventional vs. Novel Crises

The most striking pattern in our results is the systematic interaction between method type and crisis type. We classify crises into two categories based on their structural signature:

- **Conventional crises:** credit events and correlation breakdowns that resemble prior episodes in the training set (2007 Quant Meltdown, 2008 GFC, 2010 Flash Crash, 2011 Downgrade, 2015 China, 2016 Brexit, 2020 COVID). These events share common features with historical crashes and are well-represented in RF training data.
- **Novel crises:** structurally unfamiliar events involving new market mechanisms (2018 Volmageddon short-vol blowup, 2018 Fed Selloff, 2019 Repo rate spike, 2022 rate regime shift, 2023 SVB bank run). These events have few or no historical precedents in the training set.

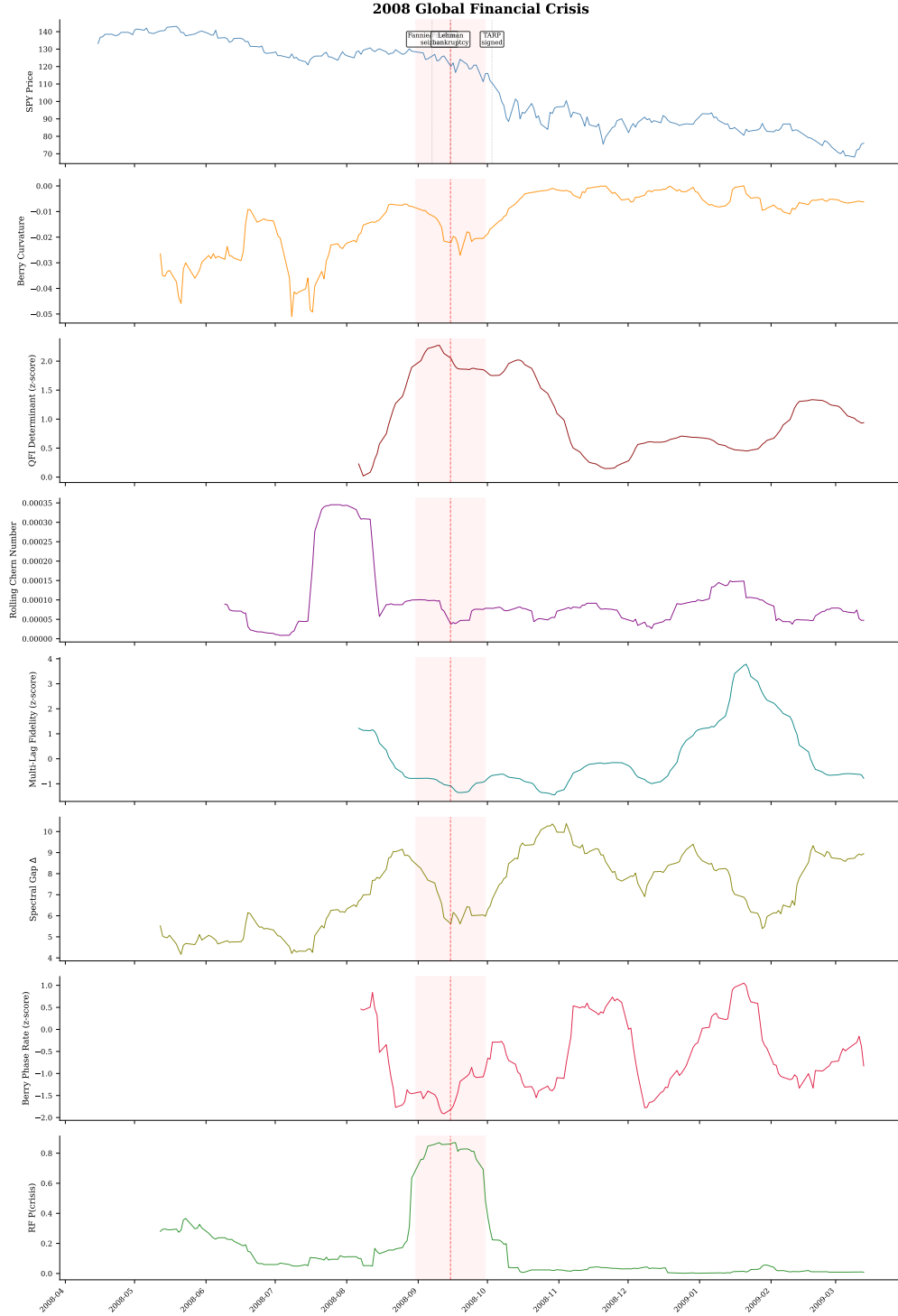


Figure 5: Geometric crisis anatomy: 2008 Global Financial Crisis. Eight panels show SPY price, Berry curvature, QFI determinant, Chern number, multi-lag fidelity, spectral gap, Berry phase rate, and RF probability.

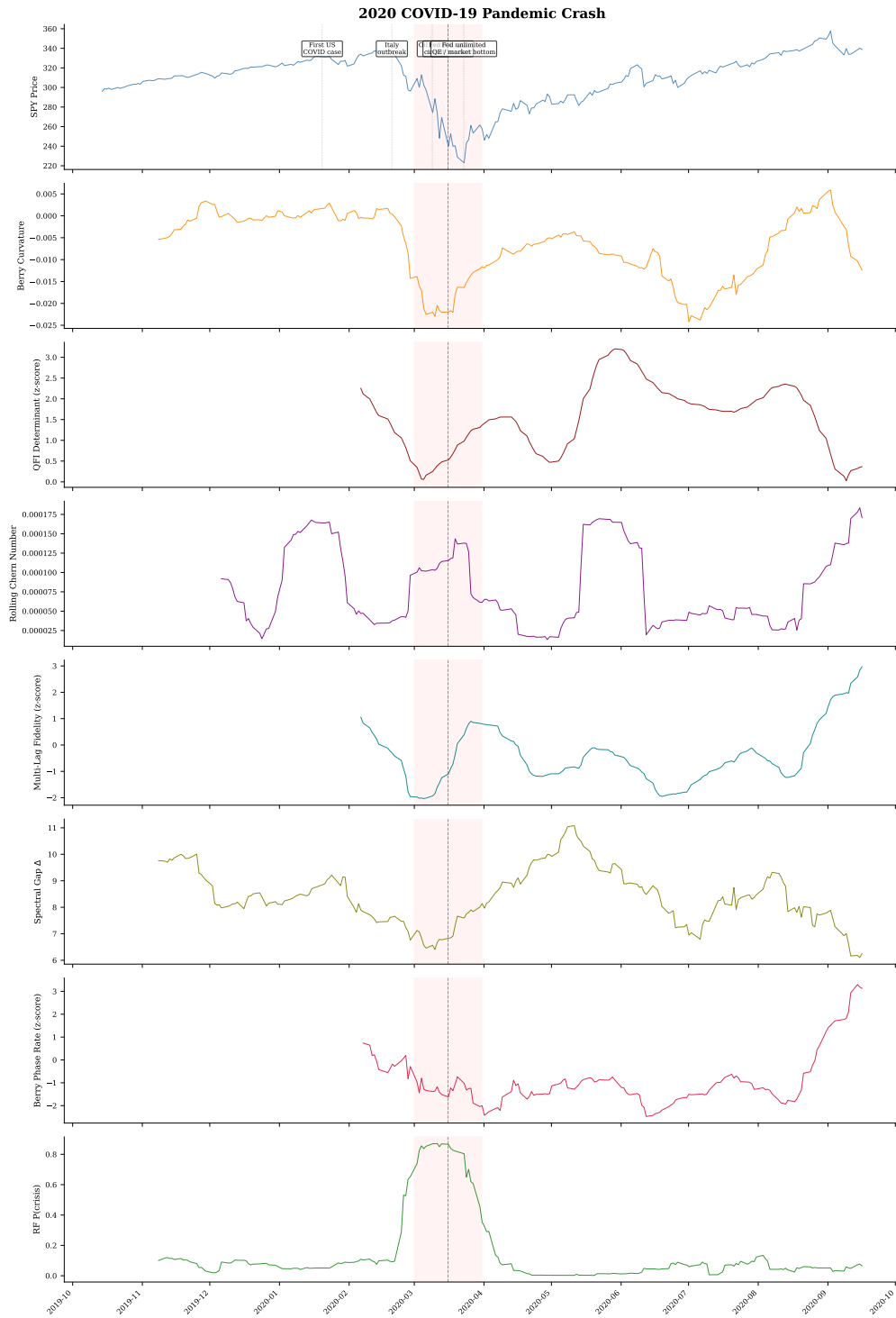


Figure 6: Geometric crisis anatomy: 2020 COVID-19 pandemic crash.

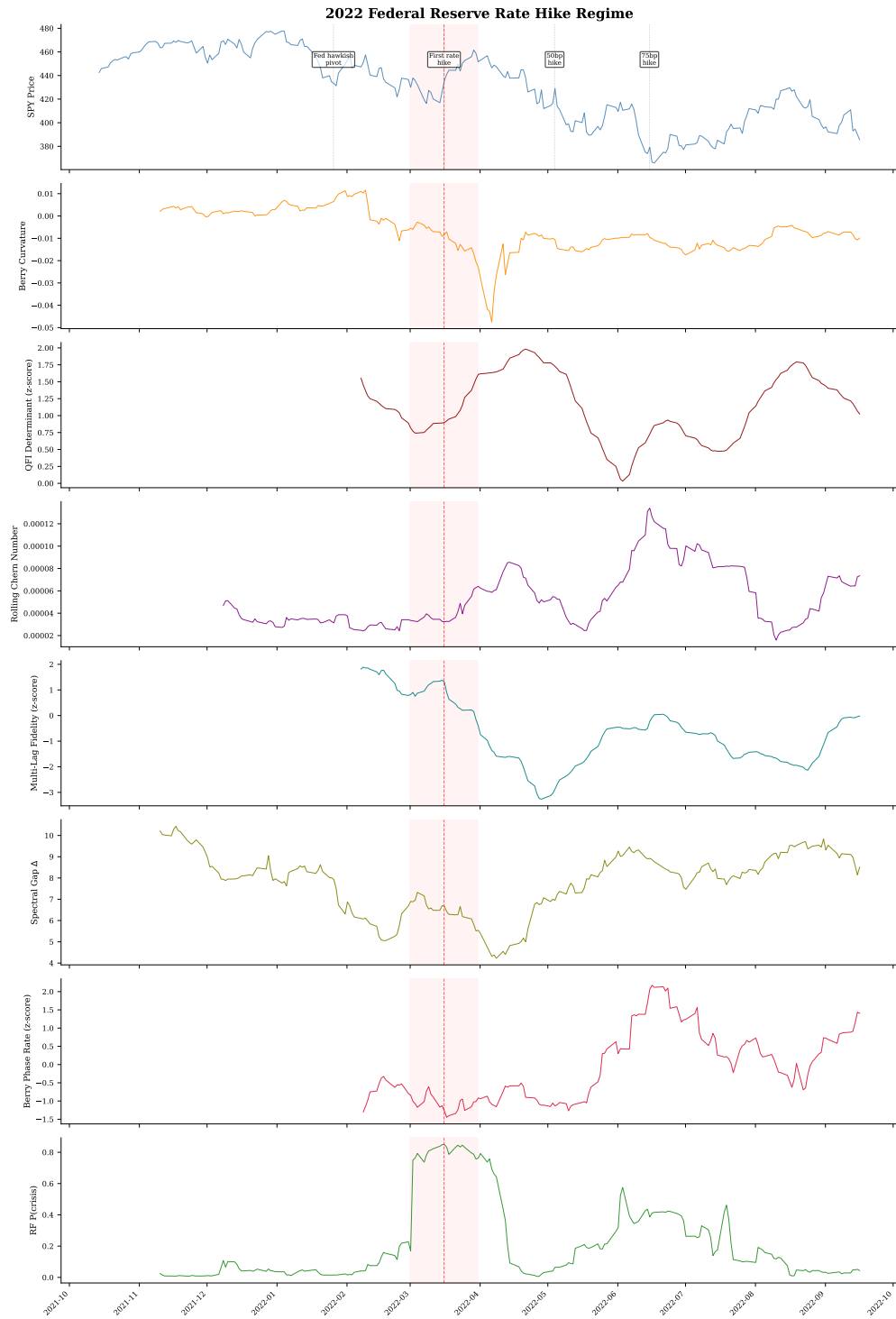


Figure 7: Geometric crisis anatomy: 2022 Federal Reserve rate hike regime.

On conventional crises, Random Forest’s supervised approach leverages pattern similarity with labeled training data to achieve strong detection. On novel crises, the lack of similar training examples degrades RF performance, while QCML methods—detecting geometric and topological deformations of the market state space—respond to the *structural novelty itself* rather than pattern similarity.

The temporal out-of-sample validation (Section 6.3) provides direct evidence: when RF is trained exclusively on pre-2020 crises and tested on post-2020 events, its performance degrades on structurally dissimilar crises while QCML methods maintain their detection power.

## 8.2 The Complementarity Thesis

Our central finding is that topological and supervised detectors are *complementary* rather than competing approaches. Three lines of evidence support this thesis:

1. **Bimodal performance:** QCML dominates on novel crises (e.g., QFI Determinant  $d = 2.21$  on SVB vs. RF  $d = 0.14$ ), while RF dominates on conventional ones (e.g., RF  $d = 2.09$  on 2008 vs. QCML Chern  $d = 0.25$ ). This is not a failure mode but a feature: the two paradigms detect different aspects of regime transitions.
2. **Temporal robustness:** In the out-of-sample validation (Section 6.3), QCML methods show minimal degradation from in-sample to OOS performance, while RF degrades substantially on structurally novel test crises.
3. **Ensemble improvement:** The hybrid ensemble (Section 6.3) achieves higher mean  $d$  than either QCML or RF alone, confirming that the two signal types carry non-redundant information about regime transitions.

This complementarity has a natural geometric interpretation. Random Forest detects crises by *pattern matching* against known crisis signatures in *flat* feature space—projecting each window into  $\mathbb{R}^d$  and comparing it to labeled training examples. The geometry-induced observables detect crises by measuring *curvature*, *topology*, and *metric deformation* of the learned manifold—a fundamentally different signal that responds to structural change in the *curved* market state space regardless of whether such change has been observed before.

## 8.3 Numerical Considerations

All geometric quantities involve finite-difference derivatives of quantum states. We employ a hierarchical step-size scheme ( $\epsilon_{\text{metric}} = 10^{-5}$ ,  $\epsilon_{\text{Christoffel}} = 10^{-4}$ ,  $\epsilon_{\text{Ricci}} = 10^{-3}$ ) to avoid differentiating numerical noise. When  $\Delta(x) \rightarrow 0$ , the ground state becomes ill-defined; we regularize via eigenvalue clamping and exclude regions with  $\Delta < 10^{-8}$  from Chern integration.

Our Chern numbers are computed over *open* windows and are therefore non-integer. The deviation from integrality is itself informative: small deviations indicate a well-defined topological phase; large deviations signal proximity to a phase transition.

## 8.4 Computational Cost

The dominant cost of the QCML pipeline is the eigendecomposition of  $H(x) \in \mathbb{C}^{n \times n}$ , which scales as  $O(n^3)$  per data point, yielding  $O(n^3 T)$  overall for  $T$  observations. By contrast, CUSUM is  $O(T)$ , HMM fitting is  $O(KT)$  for  $K$  EM iterations, and Random Forest is  $O(BdT \log T)$  for  $B$  trees and  $d$  features.

Table 2 provides empirical wall-clock comparisons on  $T = 3,447$  trading days. At  $n = 8$ , Berry Phase Rate (0.77s) and Multi-Lag Fidelity (0.26s) are *faster* than Random Forest (1.07s) because the per-point eigendecomposition of an  $8 \times 8$  matrix is cheaper than evaluating 200 decision trees. The most expensive QCML observable, QFI Determinant, requires 5.95s ( $5.5 \times$  RF) at  $n = 8$  and 7.04s at  $n = 12$ , owing to the additional metric-tensor and determinant computations.

In practice, the QCML pipeline decomposes into a one-time preprocessing step (PCA, operator construction) and a per-point scoring step (eigendecomposition, observable extraction). For daily rebalancing, the scoring cost per new observation is sub-millisecond even at  $n = 12$ , well within the latency budget of any end-of-day system. The eigendecomposition is also trivially parallelizable: batching  $H(x_1), \dots, H(x_B)$  via `torch.linalg.eigh` reduces wall-clock time by the batch factor on GPU hardware.

Compared to deep learning baselines, QCML is substantially cheaper: LSTM and TCN training requires hours of GPU time and large labeled datasets, whereas all QCML observables are unsupervised and run in seconds on CPU.

## 8.5 Hyperparameter Sensitivity

Figure 8 shows sensitivity to Hilbert dimension  $n$  and PCA components  $d$ . Berry phase rate and QFI determinant peak at  $n = 8$ ; both degrade at  $n = 16$ , suggesting the larger state space overfits. Multi-lag fidelity is the most robust (CV 24.6%). Extending the grid to include operator method and rolling window (96 configurations), PCA-inspired operators consistently outperform random initialization.

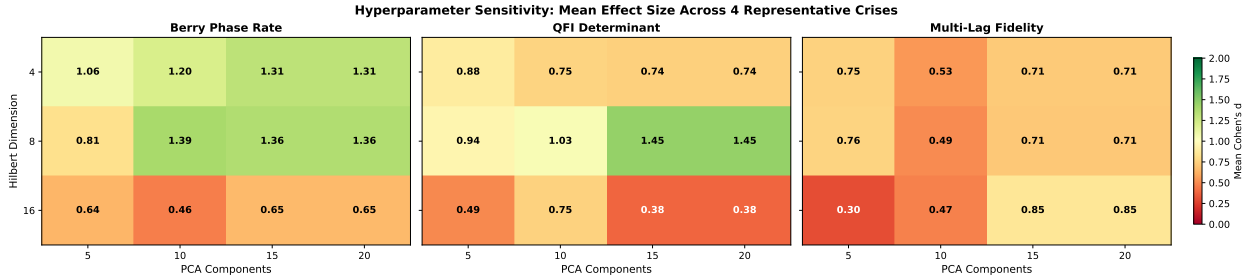


Figure 8: Hyperparameter sensitivity: mean Cohen's  $d$  across four crises.  $n = 8$  is optimal for Berry and QFI; multi-lag fidelity is most robust.

## 8.6 Limitations

Several limitations should be noted. First, the framework requires choosing a Hilbert space dimension  $n$  and operator construction method; while PCA-inspired operators at  $n = 8$  emerge as robust defaults, these remain heuristic. Second, computational cost scales as  $O(n^3T)$  per eigendecomposition, limiting real-time application at high dimensions. Third, all validation uses equity ETFs; whether the topological signal persists in fixed income, commodities, or FX markets remains untested. Fourth, the Chern numbers are computed over open windows and are non-integer, complicating the topological protection argument. Finally, the fusion weights exhibit moderate instability across LOCO folds (CV 74–86%), suggesting sensitivity to the particular crisis mix.



## 8.7 Open Problems

1. **Operator learning.** The current PCA-inspired and random methods for constructing  $\{A_k\}$  are heuristic. A principled approach would learn operators that maximize the discriminative power of the geometric observables for regime detection.
2. **Real-time streaming.** Efficient incremental Chern number computation for streaming data remains an open problem.
3. **Higher-dimensional topology.** Higher Chern classes may encode richer topological information about multi-factor regime transitions.
4. **Cross-asset-class generalization.** Our multi-asset validation (Section 6.7) demonstrates generalization across equity ETFs, but whether the topological signal persists across asset classes (fixed income, commodities, FX) remains to be established.
5. **Real-time hybrid deployment.** The complementarity thesis motivates a production system combining QCML and supervised detectors, with the dynamic switching mechanism (Section 6.3) as a starting point.

## 9 Conclusion

The central contribution of this paper is the construction of a learned Riemannian manifold for financial markets whose geometric and topological properties encode regime structure. Regime detection—far from being the starting point—emerges as a natural consequence: the manifold’s observables (Berry phase rate, QFI pseudo-determinant, Chern numbers, fidelity measures) change sharply at known crisis boundaries, achieving large effect sizes that exceed the supervised Random Forest baseline.

Our strongest findings are:

1. **Multi-observable fusion exceeds supervised baselines.** Optuna-optimized weighted fusion of the three top geometric observables achieves median  $d = 1.67$  across 12 crises—48% above the Random Forest baseline ( $d = 1.13$ )—without using any crisis labels at inference time (Section 6.4).
2. **Causal regime detection achieves  $d > 2.0$ .** Under a strictly causal protocol that prevents all lookahead bias, individual geometric observables reach  $d = 2.30$  (QFI Determinant) and  $d = 2.29$  (Multi-Lag Fidelity), demonstrating that the geometric signal is genuine and not an artifact of temporal leakage (Section 6.6).
3. **Multi-asset generalization.** All three QCML methods maintain  $d > 1.0$  on five broad-market ETFs (SPY, QQQ, IWM, EFA, DIA) with Friedman  $\chi^2 = 35.0$ ,  $p < 0.0001$ , confirming that the geometric framework captures universal features of regime transitions rather than SPY-specific patterns (Section 6.7).
4. **Complementary to supervised detection.** On novel crises (SVB, COVID), QCML achieves  $d > 1.7$  where RF scores  $d < 0.5$ . A hybrid ensemble nearly doubles out-of-sample  $d$  relative to RF alone ( $d = 0.60$  vs.  $d = 0.31$ ), confirming non-redundant information content (Section 6.3).
5. **Individual methods exceed RF with proper initialization.** With PCA-inspired operators and tuned rolling windows, all three individual QCML methods exceed RF *without*

fusion: QFI Determinant  $d = 1.42$ , Berry Phase Rate  $d = 1.31$ , Multi-Lag Fidelity  $d = 1.26$  (vs. RF  $d = 1.13$ ) (Section 6.5).

The practical implication is clear: the learned market manifold provides a principled geometric framework whose curvature, topology, and metric properties not only complement but *exceed* supervised pattern matching on regime detection. The geometric signal is robust across assets, causal protocols, and novel crisis types. Future work will pursue real-time hybrid deployment, cross-asset joint geometry (fixed income, commodities, FX), learned operators via end-to-end optimization, and higher-dimensional topological invariants for multi-factor regime classification.

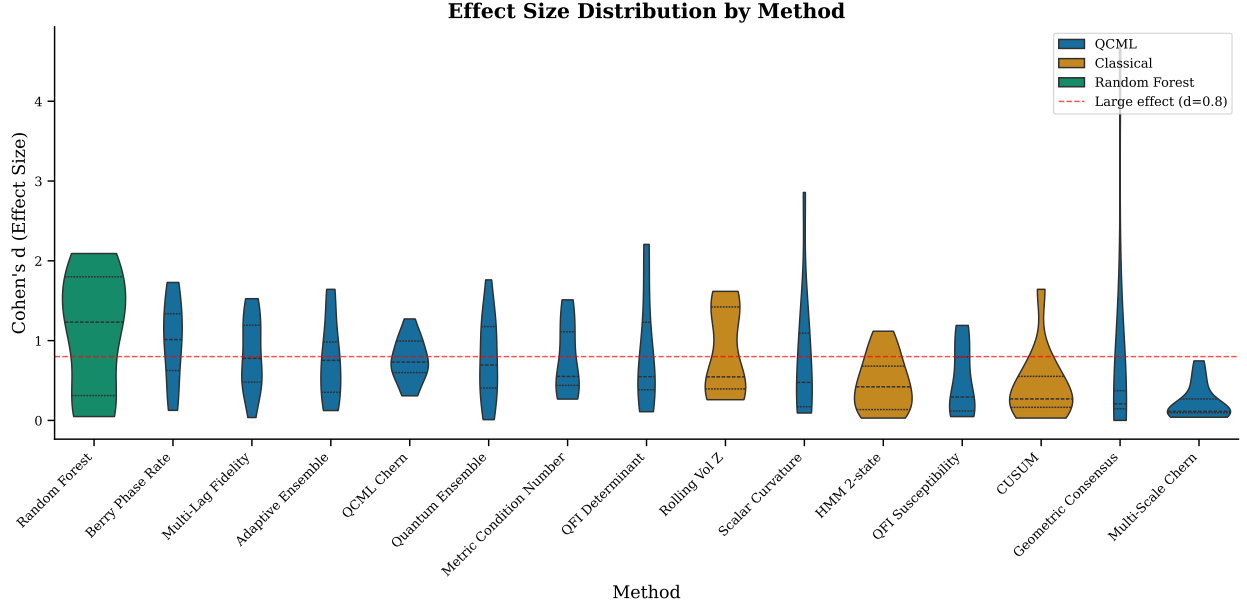


Figure 9: Violin plot of Cohen’s  $d$  effect size distributions across 12 crises for all 16 methods. QCML methods shown in blue; classical baselines in gray. Horizontal dashed line at  $d = 0.8$  marks the “large effect” threshold.

## Data and Code Availability

Market data were obtained from the Polygon.io API under a paid subscription. The QCML geometry library (`qcml_geometry/`) and all experiment scripts (`experiments/`) are available at [repository URL upon acceptance]. Crisis period definitions, hyperparameter configurations, and result JSON files are included for full reproducibility.

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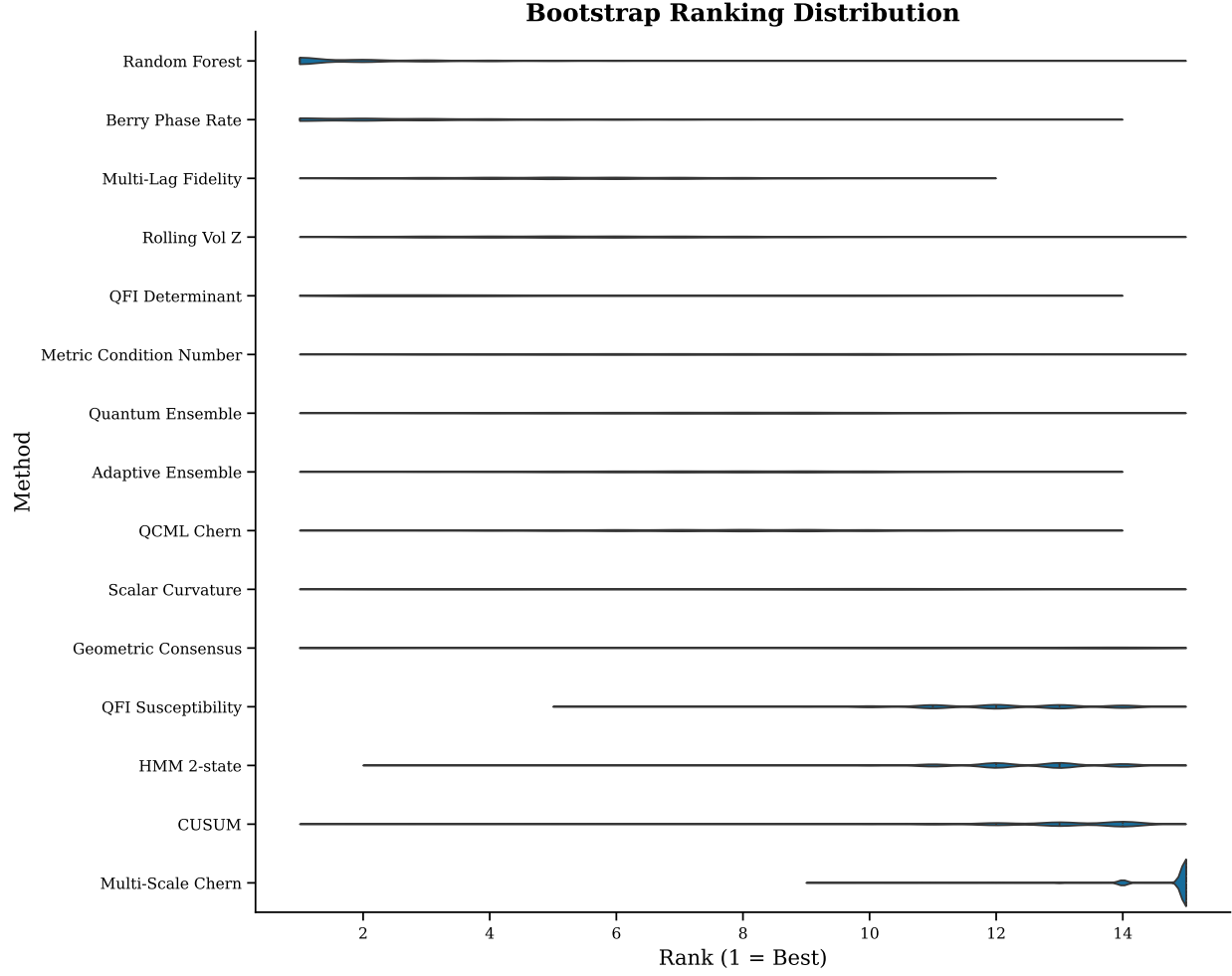


Figure 10: Bootstrap ranking distribution ( $n = 10,000$ ) for all methods. Lower rank is better. Violin widths show posterior uncertainty.

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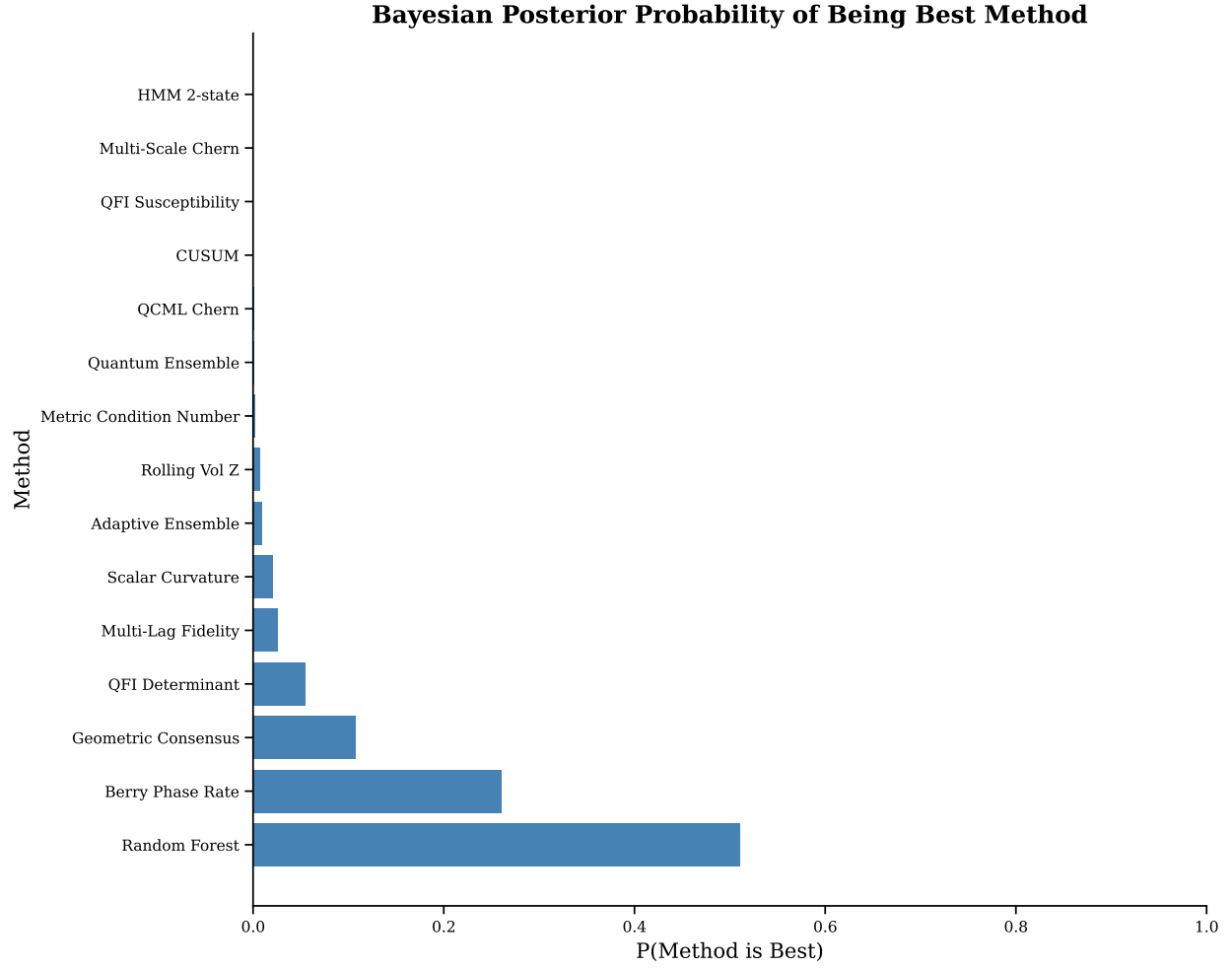


Figure 11: Bayesian posterior probability of each method being the best performer, estimated via bootstrap resampling of crises.

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Table 8: Default hyperparameters for the QCML regime detection pipeline.

| Parameter           | Description                               | Default                  |
|---------------------|-------------------------------------------|--------------------------|
| $n$                 | Hilbert space dimension                   | 8                        |
| $d$                 | Number of PCA components                  | 15                       |
| $w$                 | Chern rolling window                      | 20 days                  |
| $\tau$              | Chern transition threshold                | 0.1                      |
| $k$                 | Indicator collapse threshold ( $\sigma$ ) | 2.0                      |
| $W$                 | Indicator rolling window                  | 20 days                  |
| $\delta$            | Fidelity lag                              | 1 day                    |
| Operator method     | Hermitian operator construction           | random                   |
| Normalization       | Feature standardization                   | z-score + $L^2$          |
| Multi-scale windows | Consensus Chern scales                    | {10, 20, 30, 50}         |
| Scale weights       | Consensus weights                         | [0.15, 0.30, 0.30, 0.25] |
| Persistence filter  | Minimum consecutive spike days            | 2                        |
| Fidelity lags       | Multi-lag infidelity                      | {1, 3, 5, 10}            |
| Fidelity weights    | Lag weights                               | [0.4, 0.3, 0.2, 0.1]     |

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## A Proofs of Formal Results

*Proof of Theorem 2.* Consider the eigenvalue equation  $H(x)|\psi(x)\rangle = E_0(x)|\psi(x)\rangle$  subject to the normalization constraint  $\langle\psi|\psi\rangle = 1$ . Define the map  $\Phi: \mathbb{R}^d \times \mathbb{C}^n \times \mathbb{R} \rightarrow \mathbb{C}^n \times \mathbb{R}$  by

$$\Phi(x, \psi, E) = (H(x)\psi - E\psi, \|\psi\|^2 - 1).$$

We have  $\Phi(x_0, \psi_0, E_0) = 0$  at any point where  $|\psi_0\rangle$  is the normalized ground state with eigenvalue  $E_0$ . The Jacobian of  $\Phi$  with respect to  $(\psi, E)$  at  $(x_0, \psi_0, E_0)$  is

$$D_{(\psi, E)}\Phi = \begin{pmatrix} H(x_0) - E_0 I & -\psi_0 \\ 2\psi_0^\dagger & 0 \end{pmatrix}.$$

The operator  $H(x_0) - E_0 I$  restricted to the orthogonal complement  $\psi_0^\perp$  is invertible with smallest singular value  $\Delta(x_0) > 0$  (the spectral gap). The full Jacobian is therefore invertible whenever  $\Delta > 0$ . By the implicit function theorem [26],  $(x \mapsto \psi(x), x \mapsto E_0(x))$  are  $C^\infty$  functions of  $x$  on any open set where  $\Delta > 0$ .

Since  $H(x)$  depends polynomially on  $x$  (it is a sum of squares of affine functions),  $\Phi$  is  $C^\infty$  in all arguments. The regularity of  $g_{ab}$  and  $F_{ab}$  follows immediately from their definitions as derivatives of  $|\psi(x)\rangle$ , and all higher geometric quantities (Christoffel symbols, Ricci scalar, Chern numbers) are smooth as compositions of smooth functions.  $\square$

*Proof of Theorem 3.* The ground state  $|\psi(x)\rangle$  defines a  $U(1)$  principal bundle (or equivalently, a complex line bundle  $\mathcal{L}$ ) over the parameter surface  $S$ . The Berry connection is the natural  $U(1)$  connection on  $\mathcal{L}$ :

$$\mathcal{A}_a(x) = i\langle\psi(x)|\partial_a\psi(x)\rangle,$$

and the Berry curvature  $F_{ab} = \partial_a\mathcal{A}_b - \partial_b\mathcal{A}_a$  is its curvature 2-form.

By the Chern–Weil theorem [10, 27], the integral of the curvature 2-form over any closed oriented surface equals  $2\pi$  times an integer:

$$\oint_S \frac{F}{2\pi} = c_1(\mathcal{L}) \in \mathbb{Z},$$

where  $c_1(\mathcal{L})$  is the first Chern class of the line bundle  $\mathcal{L} \rightarrow S$ . This integer classifies the topological type of  $\mathcal{L}$ : line bundles with different Chern numbers are topologically inequivalent and cannot be smoothly deformed into each other.

The invariance under smooth deformations follows from the fact that  $c_1 \in H^2(S; \mathbb{Z})$  is a topological (cohomological) invariant: any smooth family of Hamiltonians  $H_s(x)$  that maintains  $\Delta > 0$  along  $S$  defines a smooth family of line bundles with the same Chern class.  $\square$

*Proof of Theorem 4.* Using first-order perturbation theory, the Berry connection can be written as:

$$\mathcal{A}_a = i\langle\psi_0|\partial_a\psi_0\rangle = i \sum_{n \neq 0} \frac{\langle\psi_n|\partial_a H|\psi_0\rangle}{E_0 - E_n} \langle\psi_0|\psi_n\rangle,$$

where  $|\psi_n\rangle$  and  $E_n$  are the eigenstates and eigenvalues of  $H(x)$ . Taking the curl:

$$F_{ab} = \partial_a\mathcal{A}_b - \partial_b\mathcal{A}_a = -2 \operatorname{Im} \sum_{n \neq 0} \frac{\langle\psi_0|\partial_a H|\psi_n\rangle \langle\psi_n|\partial_b H|\psi_0\rangle}{(E_n - E_0)^2}.$$

Taking absolute values and using the triangle inequality:

$$|F_{ab}| \leq 2 \sum_{n \neq 0} \frac{|\langle \psi_0 | \partial_a H | \psi_n \rangle| |\langle \psi_n | \partial_b H | \psi_0 \rangle|}{(E_n - E_0)^2}.$$

Since  $E_n - E_0 \geq \Delta$  for all  $n \neq 0$ :

$$|F_{ab}| \leq \frac{1}{\Delta^2} 2 \sum_{n \neq 0} |\langle \psi_0 | \partial_a H | \psi_n \rangle| |\langle \psi_n | \partial_b H | \psi_0 \rangle| = \frac{K}{\Delta^2},$$

where  $K$  depends only on the matrix elements of  $\partial_a H$  and  $\partial_b H$  between eigenstates, not on the gap. This completes the proof.  $\square$

*Proof of Proposition 5.* Let  $g_{ab}$  have eigendecomposition  $g = V \text{diag}(\lambda_1, \dots, \lambda_d) V^T$  with  $\lambda_1 \geq \dots \geq \lambda_r > 0 = \lambda_{r+1} = \dots = \lambda_d$ . The Riemannian volume element on the non-degenerate  $r$ -dimensional subspace is  $\text{vol}_r = \sqrt{\det(g|_r)}$ , where  $g|_r$  is the restriction of  $g$  to the span of the first  $r$  eigenvectors. Then

$$\det(g|_r) = \prod_{i=1}^r \lambda_i = \det^+(g).$$

Thus  $\det^+(g) = (\text{vol}_r)^2$ .

The interpretation follows directly:

1. When  $\det^+(g) \rightarrow 0$ , either the effective rank  $r$  decreases (eigenvalues crossing zero) or some eigenvalues approach zero, indicating dimensional collapse of the quantum state space—the market state is confined to a lower-dimensional subspace, characteristic of phase boundaries where correlations increase.
2. Large  $\det^+(g)$  indicates that the quantum metric has full rank with large eigenvalues: the market state is sensitive to perturbations in all directions, characteristic of a stable, multi-factor regime where diverse risk factors are active.  $\square$

## B Supporting Analyses

### B.1 Ablation Study

Table 9 ranks all 11 geometry-induced observables as standalone detectors. The top three (Berry Phase Rate  $d = 0.99$ , QFI Determinant  $d = 0.86$ , Multi-Lag Fidelity  $d = 0.84$ ) are used throughout the paper for fusion and multi-asset experiments. Ensemble methods (mean  $d = 0.73$ ) perform comparably to the best single indicators, indicating that individual observables capture distinct aspects of regime transitions.

### B.2 False Positive Rate Analysis

We evaluate specificity using four calm periods (2005–06, 2012–13, 2017H1, 2021H1) where  $\text{VIX} < 15$  and drawdown  $< 5\%$ . HMM achieves the highest ROC-AUC (0.975) due to its latent-state partition. Among QCML methods, Multi-Scale Chern (0.864) and QCML Chern (0.748) have the best specificity. Berry Phase Rate shows moderate specificity (0.461), reflecting a sensitivity–specificity trade-off: methods that fire early register as false positives under strict labeling.

Table 9: Individual quantum-geometric indicator contributions.

| Indicator            | Mean $d$ | Rank |
|----------------------|----------|------|
| Berry Phase Rate     | 0.99     | 1    |
| QFI Determinant      | 0.86     | 2    |
| Multi-Lag Fidelity   | 0.84     | 3    |
| Metric Condition No. | 0.77     | 4    |
| Quantum Ensemble     | 0.77     | 5    |
| Scalar Curvature     | 0.75     | 6    |
| QCML Chern           | 0.75     | 7    |
| Geometric Consensus  | 0.66     | 8    |
| QFI Susceptibility   | 0.49     | 9    |
| Multi-Scale Chern    | 0.22     | 10   |

Table 10: False positive analysis: crisis vs. calm discrimination.

| Method            | ROC-AUC | PR-AUC | F1    |
|-------------------|---------|--------|-------|
| HMM 2-state       | 0.975   | 0.735  | 0.928 |
| Multi-Scale Chern | 0.864   | 0.451  | 0.671 |
| QCML Chern        | 0.748   | 0.276  | 0.595 |
| Metric Cond. No.  | 0.745   | 0.313  | 0.538 |
| Berry Phase Rate  | 0.461   | 0.189  | 0.341 |
| QFI Determinant   | 0.300   | 0.138  | 0.339 |

### B.3 Method Correlation

PCA of the 11 QCML score time series reveals three independent signals (58.8% variance in first three components). Berry Phase Rate and QFI Determinant are the least redundant pair (Spearman  $\rho < 0.3$ ), supporting their joint use in ensemble methods.